

WEEK 6

Cable theory and dendritic computations

The brain computer

Info coming from outside $\rightarrow$ ENCODE the info
DECODE the info

Problem - Algorithm - Implementation.

Each region has a specific computational role.

Mission of the Brain: COMPUTE

• The Hubel and Wiesel Experiment

- The cells respond more when there is a line "orientation direction selective cell"



We decompose the world into parameters $\rightarrow$ colours, lines, orientation...

Early theoretical ideas for the neuron

$\rightarrow$ The McCulloch and Pitts (M&P) joint neuron - 1943



Inspired by $\left\{ \begin{array}{l} \text{All or none} \\ \text{Square E \& I} \end{array} \right.$

Assuming that a single excitatory synapse (e) can reach spike threshold and that a single inhibitory synapse (i) can veto all e-synapses then:

output (1) is generated if
(e1 or e2 or e3) and NOT i



Lord Kelvin $\rightarrow$ Mathematical model

$\downarrow$

why do we need it:

- Interpretation
- Prediction
- find key parameters.
- link biophysics to Computation. $\rightarrow$ CONCEPTUAL JUMP.

Rall $\rightarrow$ Cable Theory for Dendrites

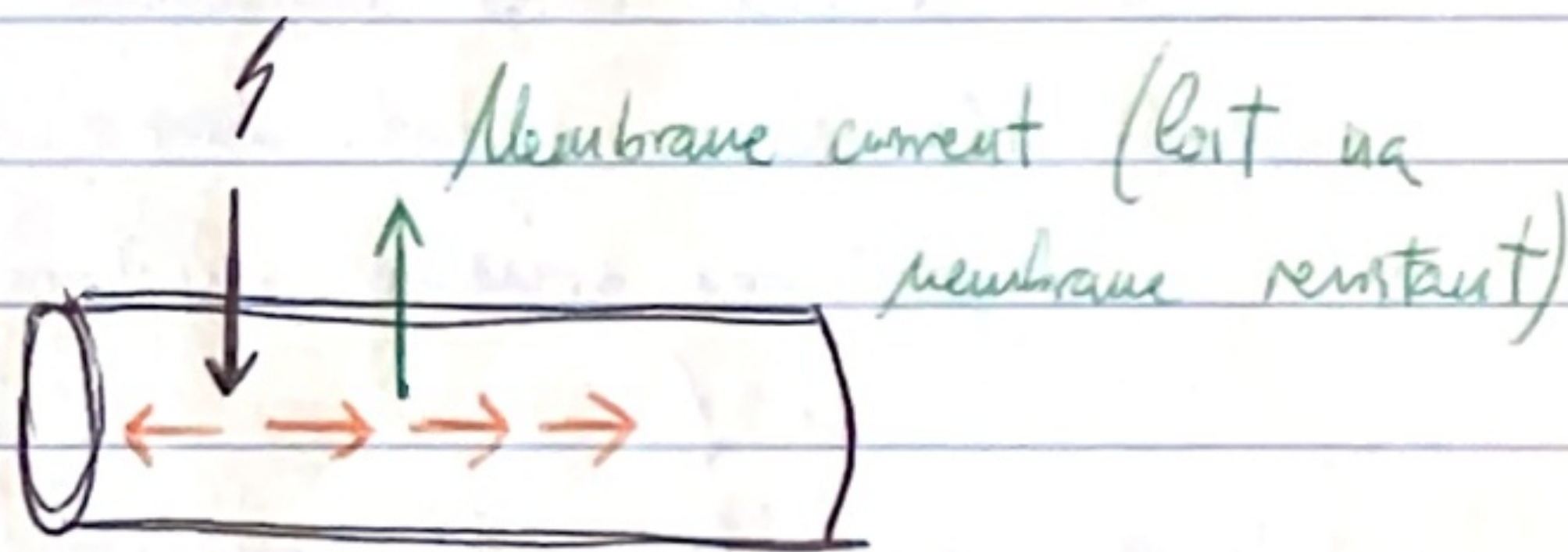
- 1) Most of the input current flows into the dendrites
- 2) Dendrites are non-isopotential electrical devices $\rightarrow$ distributed electrical system
 $\hookrightarrow$ takes time from for the action potential to reach the soma

Physiologically

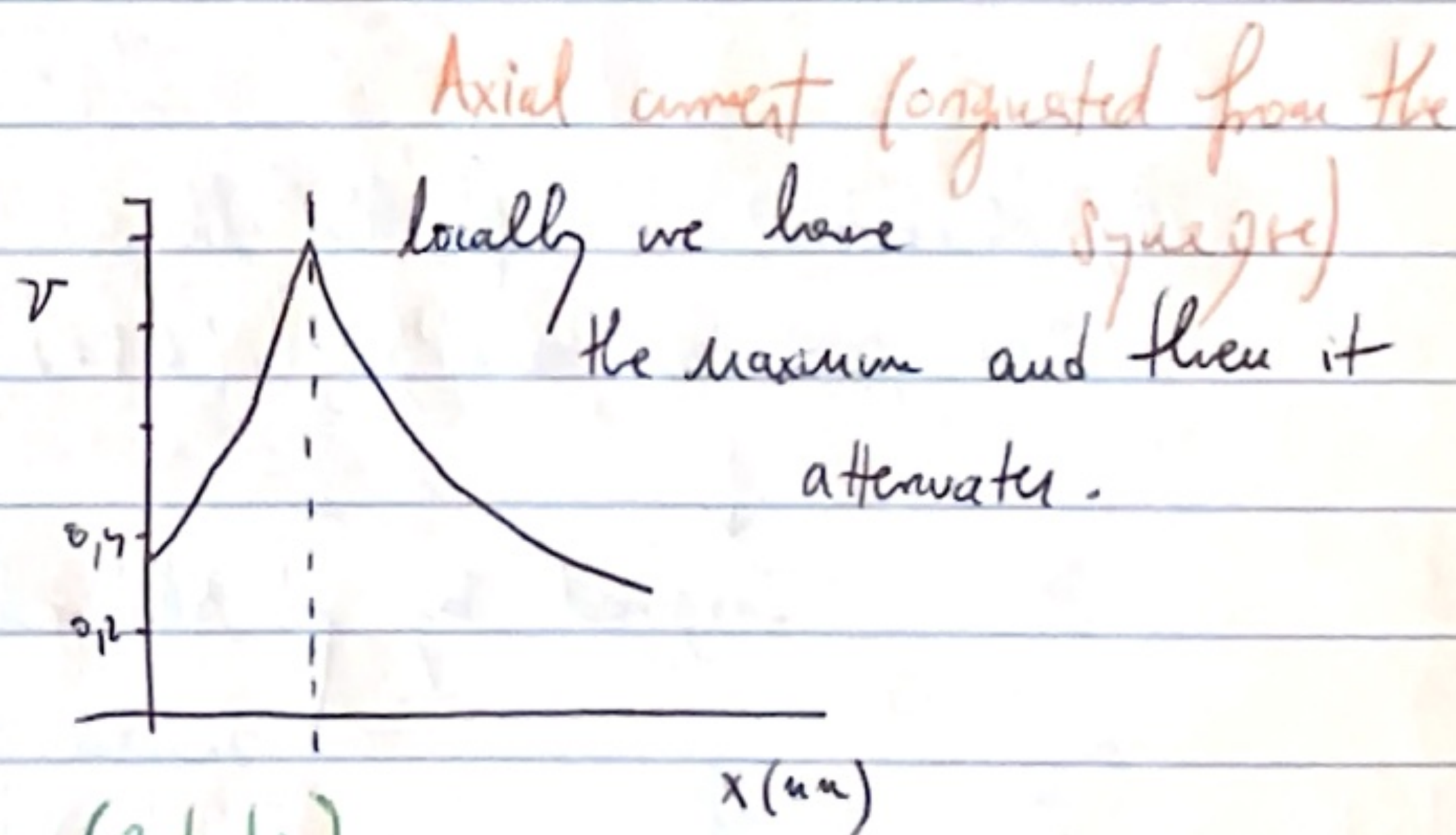
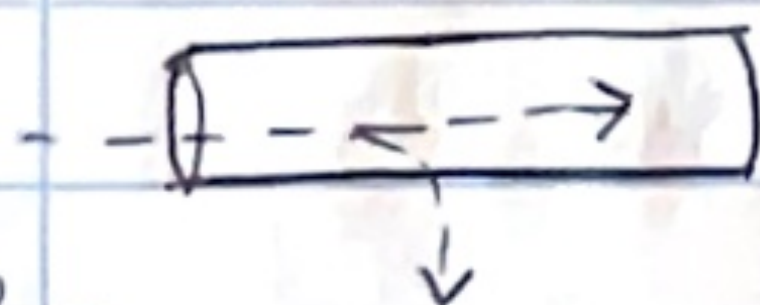


Cable

"core conductor"



the current either continues or it goes away

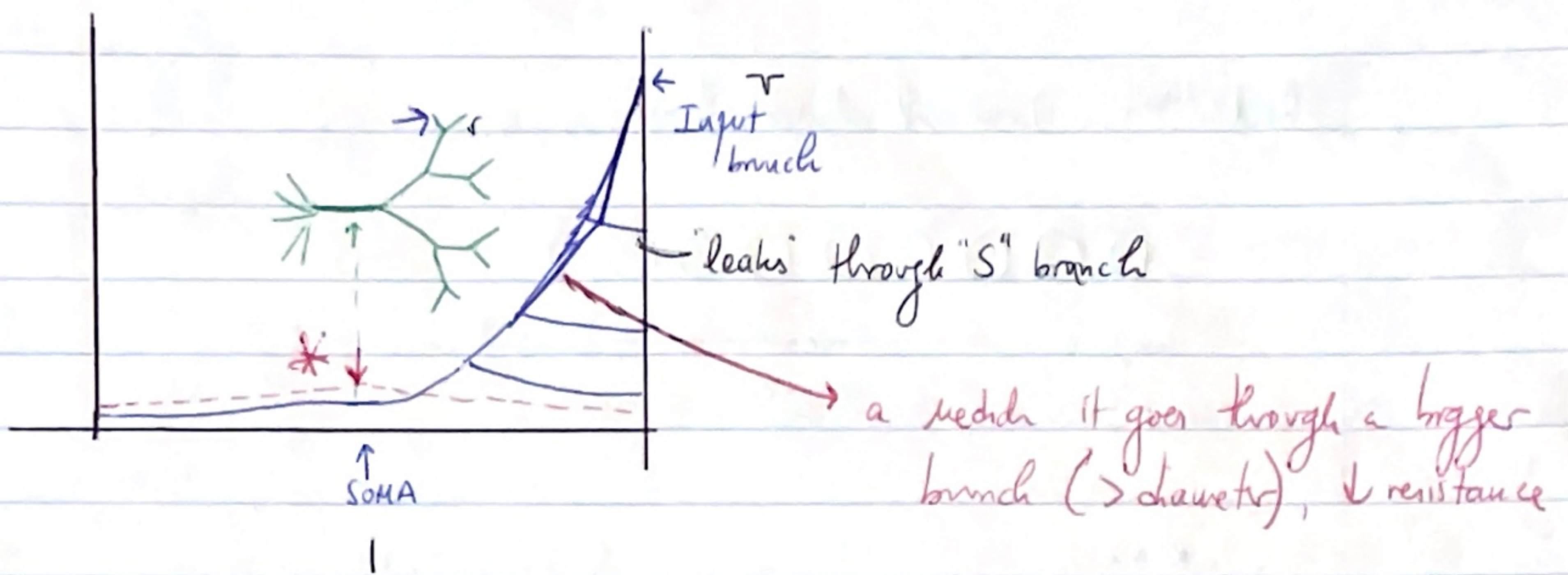


passive (electrical)

$$\left(\frac{r_m}{r_i} \right) \frac{\partial^2 V(x,t)}{\partial x^2} - \frac{r_m C_m}{\partial t} \frac{\partial V(x,t)}{\partial t} - V(x,t) = 0$$

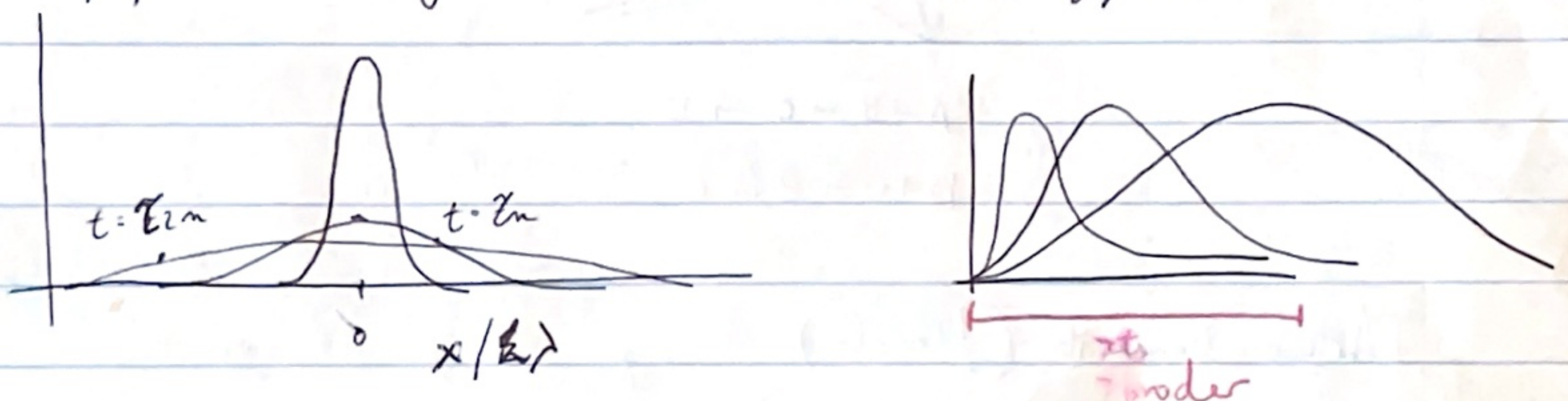
the change in the axial current

the membrane current



to the soma it reaches only a little of the PSP
but not so $\neq$ than if you inject it in the soma itself (*)
 so not so much charge is lost if we inject it in the dendrite.

each synapse has a neighbourhood that is affected strongly (locally)



- As you go away from the injection point, the curve of voltage is different
- Distal synapses are broader and delayed compared to proximal synapses.

→ You can predict where the input comes from (← here how broad is the curve)

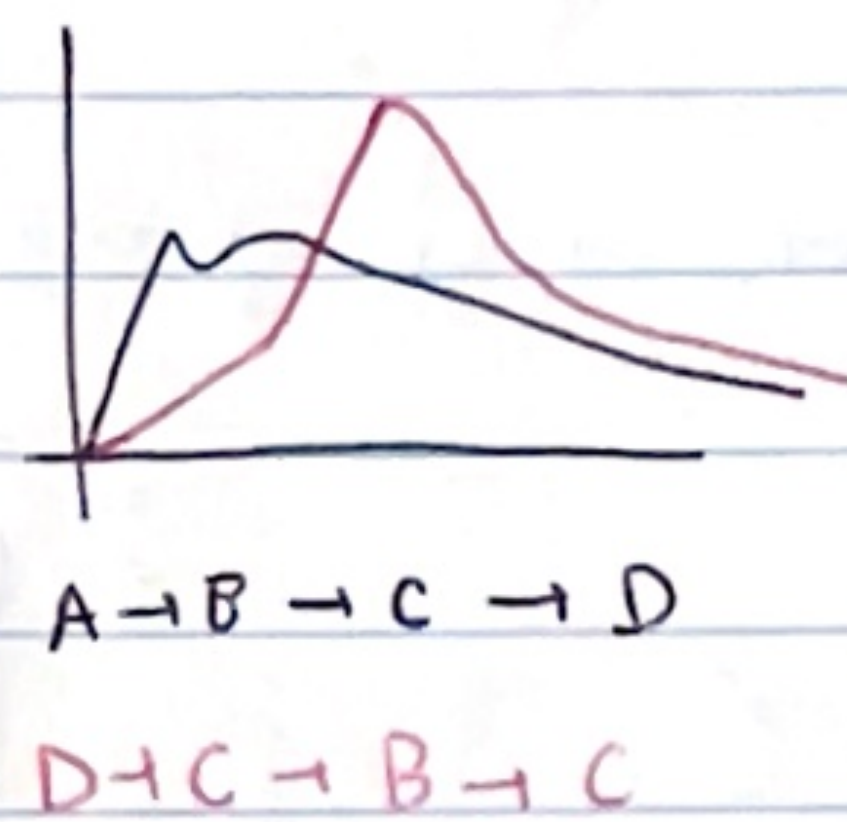
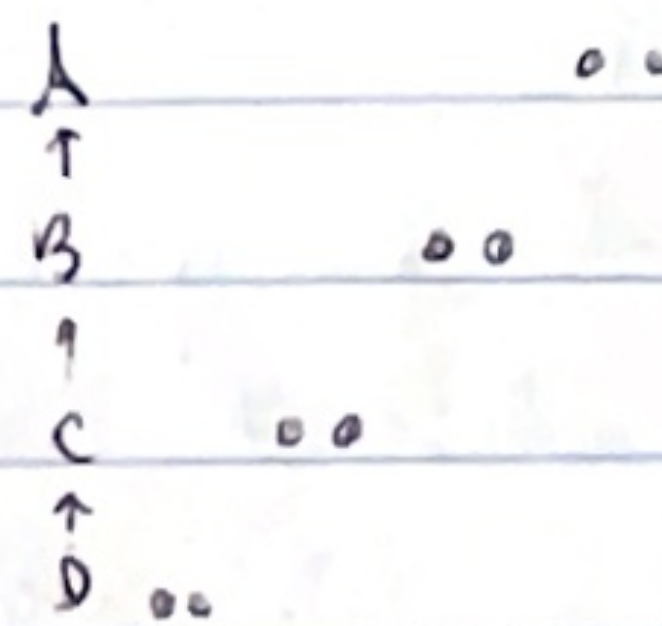
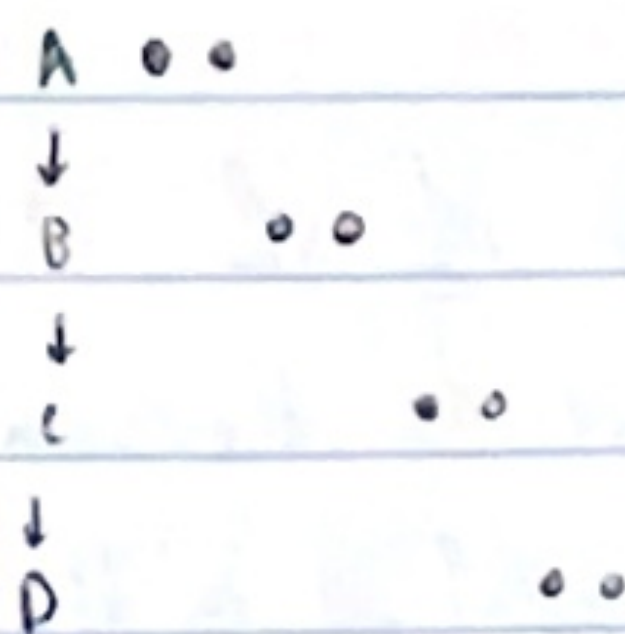
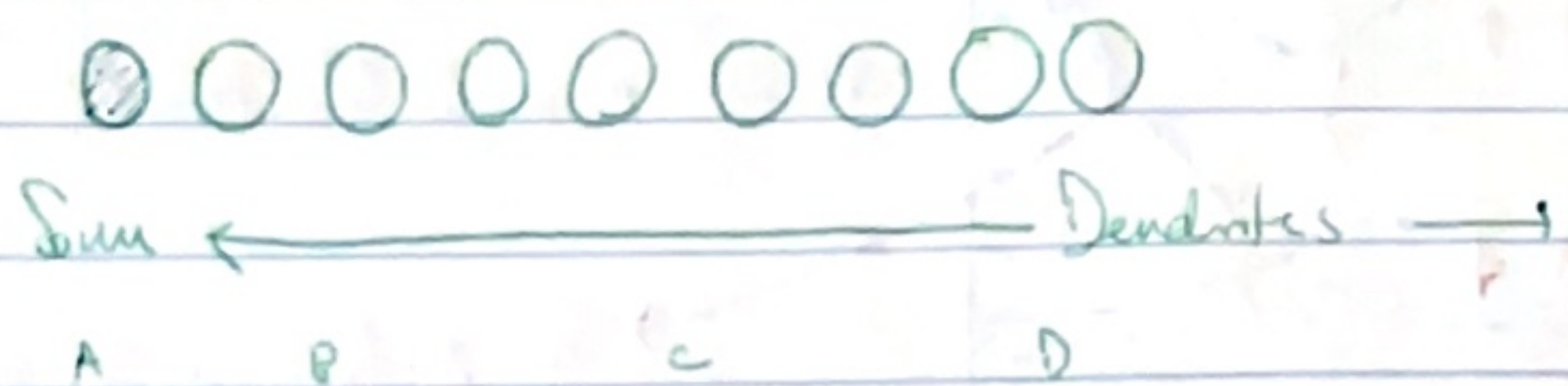
DENDRITIC COMPUTATION

- * Dendrites enable neurons as multiple functional subunits
- Dendrites work as classifiers (cluster)

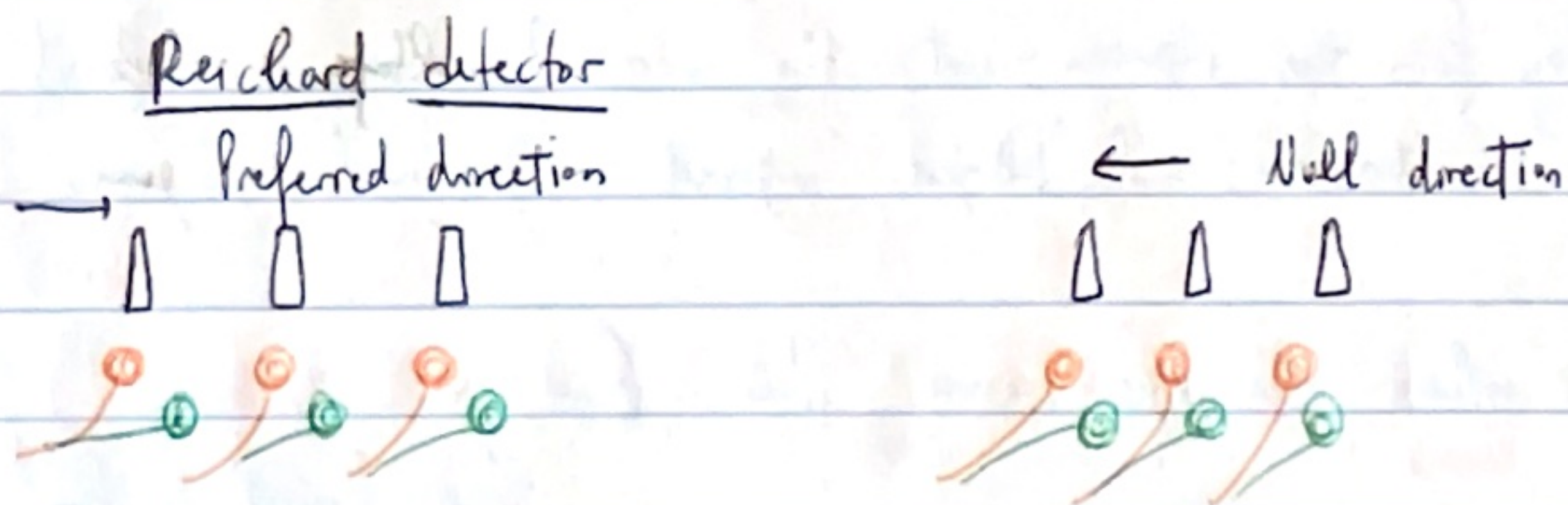
3. Mel

functional subunits } locally you integrate non-linear equations and then you sum
 regional subunits } them up.

Rall 1964 Directional selective device



Arthur Kornhuber (Munich)



1st. Inhibition Excitation 1st.

ANATOMY + FUNCTION